

Dataset Description & Data Dictionary

India's Agricultural Crop Production Analysis — Project Development Phase

Project Title	India's Agricultural Crop Production Analysis
Dataset Name	skillwallet_crop_cleaned.csv (cleaned extract)
Source	Historical Indian Crop Production Data, 1997–2021 (government agricultural statistics)
Record Count	345,407 rows
Coverage	36 states/union territories · 57 crop types · 6 growing seasons · 25 years (1997–2021)
Prepared By	Divya Kauluri (Team Project — completed individually; the team lead and other assigned members did not contribute to the work)
Date	02/07/2026

Important Note on This Document

This document provides the schema, field definitions, and cleaning methodology for the dataset used throughout this project. The original raw CSV file itself was processed locally during development and was not uploaded into this conversation, so it cannot be repackaged or exported here as a literal data file. If you upload the raw or cleaned CSV, it can be attached directly alongside this description in a future package.

1. Data Dictionary

Field	Type	Description
Country	Text	Fixed value 'India'; retained for potential future multi-country comparison.
State_Name	Text (Dimension)	Name of the Indian state or union territory where the crop was recorded.
District_Name	Text (Dimension)	District-level granularity within each state (available for future drill-down).
Crop_Year / Crop_Year_Numeric	Text / Number	Agricultural year of the record; the numeric version drives all trend charts and the year-range filter (1997–2021).
Season	Text (Dimension)	Growing season — Kharif, Rabi, Autumn, Summer, Winter, or Whole Year — used for seasonal comparison.
Crop	Text (Dimension)	Crop name (e.g., Rice, Wheat, Cotton(lint), Soyabean, Bajra, Maize, Jowar, Gram, Groundnut, Rapeseed & Mustard, and other crops).
Area / Area (M)	Number	Cultivated area for the record, in original units and in millions (M) for readable axis scales.
Area_Units	Text	Unit of the Area field (Hectare).
Production / Production (M)	Number	Total quantity produced, in original units and in millions (M).
Production_Units	Text (Dimension)	Unit of measure for

		Production — Tonnes, Bales, or Nuts — which differ by crop and must be interpreted separately.
Production_Normalized	Number (Calculated)	Production converted to a common tonnage-equivalent basis where feasible, added to reduce distortion from mixed units.
Yield	Number (Calculated)	Production divided by Area, representing productivity per unit of land; averaged for the yield-trend line chart.

2. Cleaning & Preparation Summary

Full step-by-step preprocessing detail is documented in ‘1. Preprocessing Steps and Business Questions.docx/pdf’ in this same folder. In summary: duplicate and blank rows were removed, state and crop names were standardised, Crop_Year_Numeric and Yield were derived, Production_Normalized was calculated to account for mixed units, and Area/Production were scaled to millions for chart readability.

3. Known Limitations

- 2021 figures are partial-year, since year-end government reporting was incomplete at the time of data collection.
- Production_Units differ by crop (Tonnes, Bales, Nuts); comparisons across crops or states should always be read alongside the Production Distribution by Unit Type chart.
- District-level data exists in the schema but was not used for district-level charts in this phase of the project; it is flagged as a Future Scope item.